MISKATONIC UNIVERSITY

Arkham, Massachusetts · Established 1797

Course Unit Description

Course Unit	CNC Machining			**Field of study**	Mechanical Engineering		
Degree programme	Higher Technical Diploma in Mechanical Production			**School**	School of Engineering and Applied Technology		
Academic Year	2026/2027	**Year of study**	1	**Level**	Short cycle	**ECTS credits**	8.0
Type	Semestral	**Semester**	2	**Code**	MU-MPT-1112		

Workload (hours)	216	**Contact hours**	**T**	15	**TP**	-	**PL**	75	**TC**	-	**S**	-	**E**	-	**OT**	5	**O**	-

T: Lectures; TP: Lectures and problem-solving; PL: Problem-solving, project or laboratory; TC: Fieldwork; S: Seminar; E: Placement; OT: Tutorial; O: Other

Name(s) of lecturer(s)	Walter Grimsby, Ana Lucia Moreira

Learning outcomes and competences

At the end of the course unit the learner is expected to be able to:

1. work out from a technical drawing what a part must look like, including tolerances, surface finish and material, and plan the order of operations needed to make it
2. set up a CNC lathe and a three-axis machining centre for a new job: clamping the workpiece, loading and measuring the tools, and setting the work offsets
3. write and edit simple part programs in ISO G-code and check them in simulation before running them on the machine
4. run a job and correct it while it is running, changing feeds, speeds and tool offsets when the cut, the surface or the measured dimensions are not as expected
5. check finished parts against the drawing with calipers, micrometers, bore gauges and a height gauge, and record the results
6. keep the machine and the work area safe and in good order

Prerequisites

Before the course unit the learner is expected to be able to:

1. use basic hand tools and conventional workshop machines safely
2. carry out basic trigonometric and unit calculations

Course contents

This unit takes students from a drawing to a finished part on CNC lathes and machining centres. Students learn how to read what the designer asks for, how to prepare the machine and the program, how to run the job and how to make sure the part is right. Almost all of the time is spent in the workshop, working on parts of increasing difficulty.

Course contents (extended version)

1. Reading the drawing
   - views, sections and dimensioning
   - dimensional and geometric tolerances
   - surface finish symbols
   - choosing the sequence of operations
2. The CNC machine
   - machine axes, coordinate systems and reference points
   - the control panel and operating modes
   - safety in the CNC workshop
3. Preparing a job
   - holding the workpiece: chucks, vices and fixtures
   - tool selection and tool holders
   - measuring tool length and radius
   - setting work offsets
4. Programming
   - structure of a part program
   - linear and circular interpolation
   - tool radius compensation
   - canned cycles for drilling, turning and pocketing
   - simulation and dry runs

5. Cutting conditions
   - cutting speed, feed and depth of cut
   - tool wear and its effect on the part
   - dealing with vibration, poor finish and dimensions drifting out of tolerance
6. Measuring and inspection
   - calipers, micrometers, bore gauges and height gauges
   - measurement errors and how to avoid them
   - first-article inspection and inspection records

Recommended reading

1. CNC Programming Handbook, Third Edition. Peter Smid, Industrial Press, 2008.
2. Technology of Machine Tools. Steve F. Krar, Arthur R. Gill and Peter Smid, McGraw-Hill, latest edition.
3. Machinery's Handbook. Erik Oberg, Franklin D. Jones, Holbrook L. Horton and Henry H. Ryffel, Industrial Press, latest edition.
4. Fundamentals of Dimensional Metrology. Connie L. Dotson, Cengage Learning, latest edition.

Teaching and learning methods

Short lectures (T) introduce each topic, but the unit is taught mainly in the workshop (PL), in groups of at most eight students per lecturer. Each student produces a series of parts of increasing difficulty, starting from simple turned parts and ending with a part that needs both turning and milling operations.

For each part, students prepare a process sheet and the program, have them approved by the lecturer, machine the part and fill in an inspection report. Tutorials (OT) are used to review programs and inspection results individually.

Assessment methods

1. Alternative 1 (Regular, Student Worker) (Final, Supplementary)
   - Workshop portfolio: process sheets, programs and inspection reports for all parts - 30%
   - Final practical test: producing a part to drawing within a set time - 50% (the minimum grade of 10 out of 20 points is required)
   - Written test - 20%
2. Alternative 2 (Regular, Student Worker) (Special)
   - Final practical test - 70%
   - Written test - 30%

Language of instruction

1. English

Electronic validation

Course coordinator	Programme director	Head of School	Pro-Rector for Teaching
Walter Grimsby	Peter Lindahl	Rosalind Carver	Henrik Aaltonen
01-10-2026	05-10-2026	09-10-2026	15-10-2026